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B.Tech

Web Technology

KCS-602

With Notes

UNIT-5

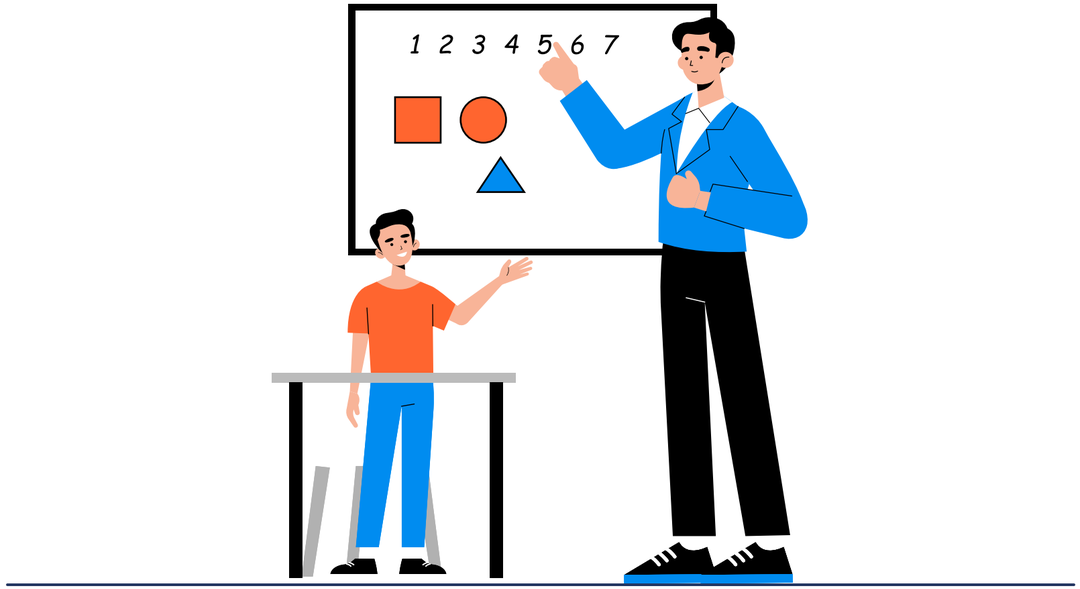
Servlets & Java Server Pages (JSP)

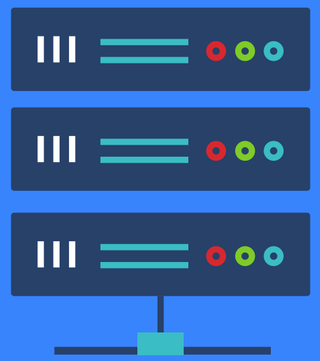
(in one video)

AKTU Exam

Topics to be covered...

Servlet
Servlet life cycle
Common Gateway Interface (CGI)
Cookies
Java Server Page (JSP)
JSP Important tags
JSP Architecture
Tomcat Server
Implicit objects available in JSP
Standard actions in JSP
JSP Directives
Happy Ending!

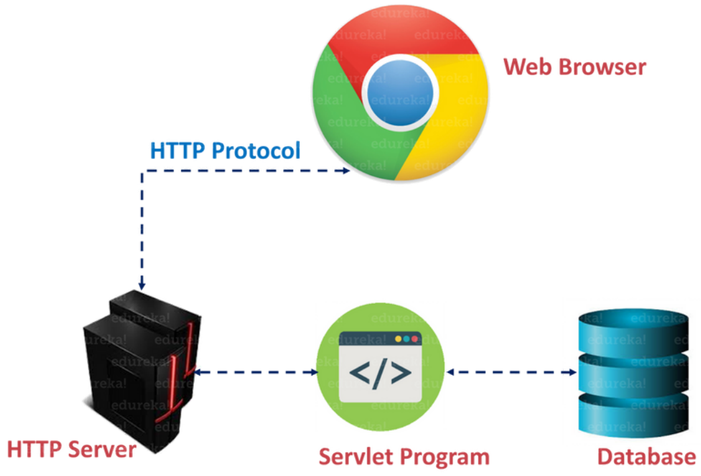




Servlet

Servlet

- Servlets are the Java programs that run on the Java-enabled web server.
- They are used to handle the request obtained from the webserver, process the request, produce the response, then send a response back to the webserver.
- Servlet also called servlet.
- Used in the cookies and session tracking.



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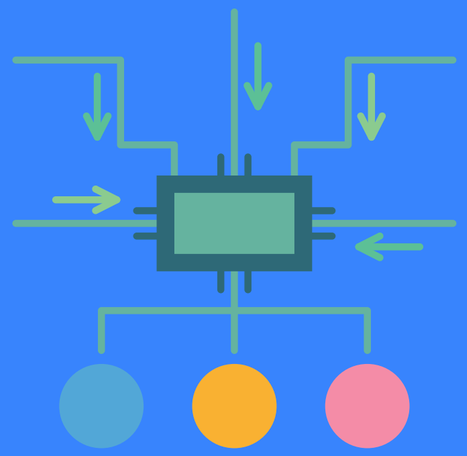
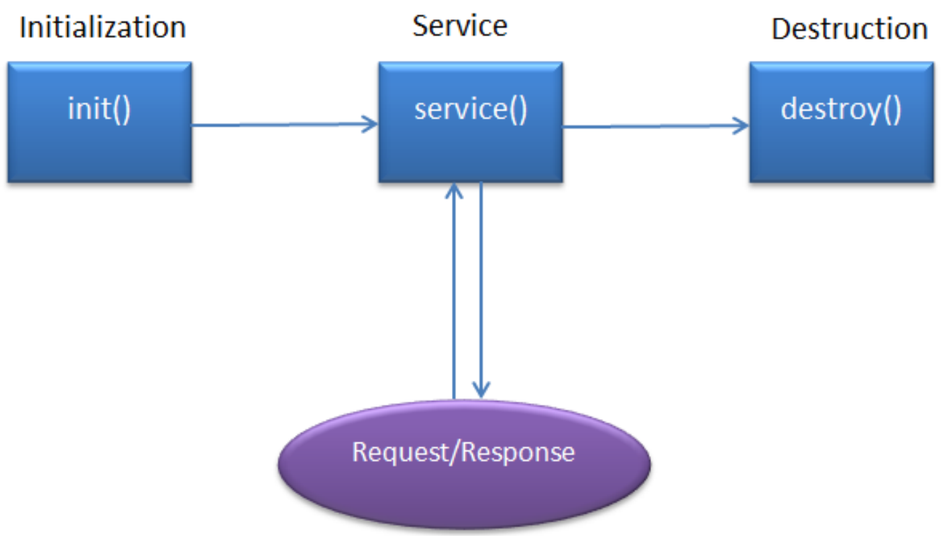


Servlet life cycle

Servlet life cycle

- init()
- service()
- destroy()



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Servlet life cycle

init()

- The **init()** method is called only once.
- It is called only when the servlet is created, and not called for any user requests afterwards.
- It is used for one-time initializations, just as with the **init()** method of applets.
- The **init()** method must complete successfully before the servlet can receive any requests.

```
public void init() throws ServletException {
    // Initialization code...
}
```

Servlet life cycle

service()

- The **service()** method is called only after the **init()** method.
- It is the main method to perform the actual task.
- The servlet container calls the **service()** method to handle requests coming from the client.
- The **service()** method checks the HTTP request type (GET, POST, PUT, DELETE, etc.)

```
public void service(ServletRequest request, ServletResponse response)
   throws ServletException, IOException {
}
```

Servlet life cycle

destroy()

- The **destroy()** method is called only once at the end of the life cycle of a servlet.
- It gives your servlet a chance to close database connections.
- After the **destroy()** method is called, the servlet object is marked for garbage collection.

```
public void destroy() {
    // Finalization code...
}
```

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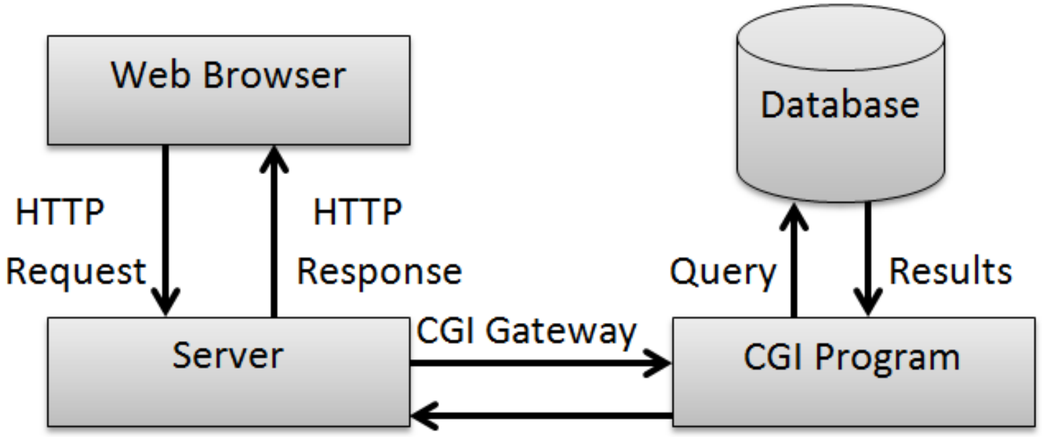
Common Gateway Interface (CGI)

Common Gateway Interface (CGI)

- CGI is a standard way for a Web server to pass a Web user's request to an application program and to receive data back to forward to the user.
- CGI used to provide dynamic content to the user.
- The World Wide Web Consortium (W3C) defined the Common Gateway Interface (CGI) and also defined how a program interacts with a Hyper Text Transfer Protocol (HTTP) server.



Java servlets advantages over CGI

- **Platform independence:** Java servlet are pure Java program and it can run on any servlet enabled web server.

- **Safety:** Memory management, exception handling.

- **Secure:** Servlets are server side components, so it inherit the security provided by the web server.
- **Performance**
- **Extensibility**

Cookies

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Cookies

- Cookies are small files which are stored on a user's computer.
- When you select your language in which you want to view your website, the website would save the information in a document called a cookie on your computer, and the next time when you visit the website, it would be able to read a cookie saved earlier. That way the website could remember your language and let you view the website in your preferred language without having to select the language again.
- A cookie can contain any type of information such as the time when you visited the website, the items that you added into your shopping basket, all the links you clicked in website, etc.,

Cookies

Types of Cookies:

- **Session Cookies –** These are mainly used by online shops and allows you to keep items in your basket when shopping online. These cookies expire after a specific time or when the browser is closed.

- **Permanent Cookies –** These remain in operation, even when you have closed the browser. They remember your login details and password so you don't have to type them in every time you use the site. It is recommended that you delete these type of cookies after a specific time.

- **Third-Party Cookies –** These are installed by third parties for collecting certain information. For example: Google Maps.

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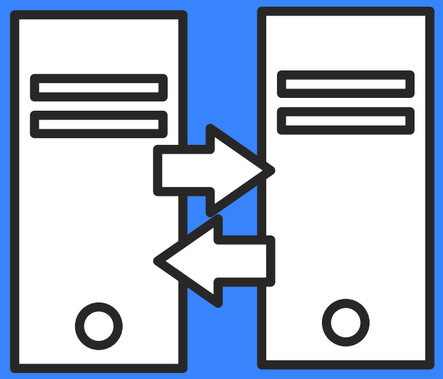

Java Server Page (JSP)

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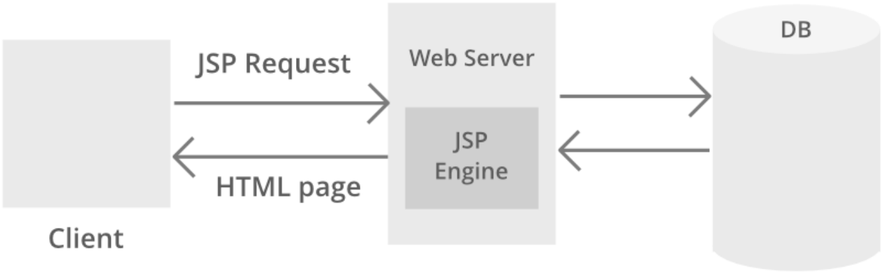
Java Server Page (JSP)

- JSP technology is used to create web application just like Servlet technology.
- It supports both scripting and element based dynamic content and allow developers to create their own tag libraries
- A JSP page consists of HTML tags and JSP tags.
- JSP tags are reusable. So, the pages run faster.
- Java Server Pages often serve the same purpose as programs implemented using the Common Gateway Interface (CGI). But JSP offers several advantages in comparison with the CGI.



JSP Important tags

Declarative Tag

```
<%!

  //variables
  //methods

%>
```

Script Tag

```
<%

  //Java source code

%>
```

Expression Tag

<%= Statement %>

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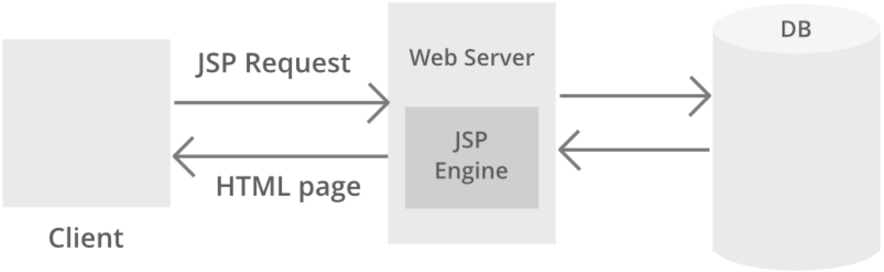
JSP Architecture

JSP Architecture

- JSP architecture is a 3 tier architecture.
- The client is the web browser or application on the user side.
- It is a server-side technology for creating web applications.
- SP tags are used to insert JAVA code into HTML pages.
- It is an advanced version of Servlet Technology.
- Create dynamic and platform-independent web pages.



JSP Architecture

- **Step1:** Client navigates to a file the .jsp extension and the browser initiates an HTTP request to the webserver.
- **Step2:** If the compiled version of JSP exists in the web server, it returns the file. Otherwise, the request is forwarded to the JSP Engine.
- **Step 3:** The JSP Engine loads the JSP file and translates the JSP to Servlet(Java code). This process is called translation.
- **Step4:** The JSP engine compiles the Servlet to an executable .class file. It is forwarded to the Servlet engine. This process is called compilation phase.
- **Step 5:** The .class file is executed by the Servlet engine which is a part of the Web Server. The output is an HTML file. The Servlet engine passes the output as an HTTP response to the webserver.
- **Step 6:** The web server forwards the HTML file to the client's browser.

JSP Architecture

```
<html>
<body>
<form action="welcome.jsp">
<input type="text" name="uname">
<input type="submit" value="go"><br/>
</form>
</body>
</html>
```

```
<html>
<body>
<%
String name=request.getParameter("uname");
out.print("welcome "+name);
%>
</form>
</body>
</html>
```

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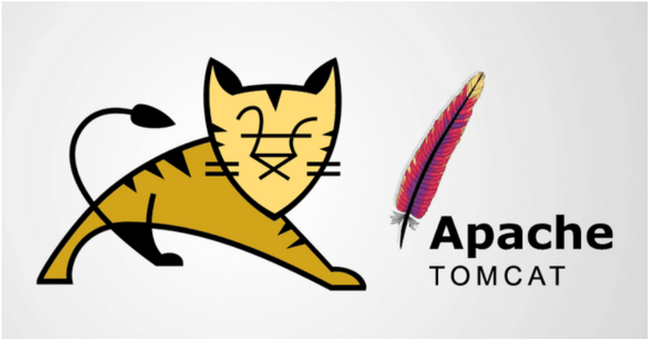
Tomcat Server

Tomcat Server

- It is an open-source Java servlet container that implements many Java Enterprise Specs such as the Websites API, Java-Server Pages and last but not least, the Java Servlet.
- Complete name of Tomcat is "Apache Tomcat".
- Static website- HTTP server
- Dynamic website- servlet
- HTTP server + servlet = become a single server know as tomcat server.
- We can say servlet container



Implicit objects available in JSP

Implicit objects available in JSP

- There are 9 jsp implicit objects.
- These objects are created by the web container that are available to all the jsp pages.

Object	Type
out	JspWriter
request	HttpServletRequest
response	HttpServletResponse
config	ServletConfig
application	ServletContext
session	HttpSession
pageContext	PageContext
page	Object
exception	Throwable

Implicit objects available in JSP

1) JSP out object:

- For writing any data to the buffer.
- It is the object of JspWriter.
- In case of servlet you need to write: PrintWriter out=response.getWriter();

```
1. <html>
2. <body>
3. <% out.print("Today is:"+java.util.Calendar.getInstance().getTime()); %>
4. </body>
5. </html>
```

Implicit objects available in JSP

2) JSP request object:

- **Type:** HttpServletRequest
- It can be used to get request information such as parameter, header information, server name, server port, content type etc.
- It can also be used to set, get and remove attributes from the jsp request scope.

```
<form action="welcome.jsp">
<input type="text" name="uname">
<input type="submit" value="go"><br/>
</form>
```

```
<%
String name=request.getParameter("uname");
out.print("welcome "+name);
%>
```

Implicit objects available in JSP

3) JSP response object:

- **Type:** HttpServletResponse

```
<%
response.sendRedirect("http://www.google.com");
%>
```

4) JSP config object:

- **Type:** type ServletConfig
- It is used to get initialization parameter from the web.xml file.

```
<%
out.print("Welcome "+request.getParameter("uname"));

String driver=config.getInitParameter("dname");
out.print("driver name is="+driver);
%>
```

Implicit objects available in JSP

5) JSP application object:

- **Type:** ServletContext.
- The instance of ServletContext is created only once by the web container when project is deployed on the server.
- This initialization parameter can be used by all jsp pages.

6) JSP session object:

- **Type:** HttpSession
- The Java developer can use this object to set,get or remove attribute or to get session information.

7**) JSP pageContext object:**

- **Type:** PageContext class.
- The pageContext object can be used to set,get or remove attribute from one of the following scopes: page, request, session, application.

Implicit objects available in JSP

8) JSP page object:

- **Type:** Object class.
- This object is assigned to the reference of auto generated servlet class.

9) JSP exception object:

- **Type:** java.lang.Throwable class
- This object can be used to print the exception.
- But it can only be used in error pages.

```
<%@ page isErrorPage="true" %>
<html>
<body>

Sorry following exception occured:<%= exception %>

</body>
</html>
```

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Standard actions in JSP

Standard actions in JSP

There are 12 types of Standard Action Tags in JSP.

- jsp:useBean
- jsp:include
- jsp:setProperty
- jsp:getProperty
- jsp:forward
- jsp:plugin
- jsp:attribute
- jsp:body
- jsp:text
- jsp:param
- jsp:attribute
- jsp:output

Standard actions in JSP

jsp:useBean

- This action name is used when we want to use beans in the JSP page.
- With this tag, we can easily invoke a bean.

```
<jsp:useBean id="" class="" />
```

jsp:include

- It also used to insert a jsp file into another file, just like including Directives.
- It is added during request processing phase

```
<jsp:include page="page URL" flush="true/false">
```

jsp:setProperty

- This property of standard actions in JSP is used to set the property of the bean.
- We need to define a bean before setting the property

Standard actions in JSP

jsp:getProperty

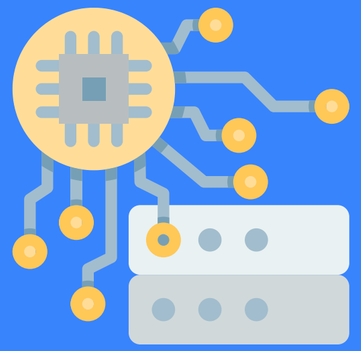
- This property is used to get the property of the bean.
- It converts into a string and finally inserts into the output.

jsp:forward

- It is used to forward the request to another jsp or any static page.
- Here the request can be forwarded with no parameters or with parameters.

jsp:plugin

- It is used to introduce Java components into jsp, i.e., the java components can be either an applet or bean.
- It detects the browser and adds <object> or <embed> JSP tags into the file



JSP Directives

JSP Directives

The jsp directives are messages that tells the web container how to translate a JSP page into the corresponding servlet.

There are three types of directives:

- page directive
- include directive
- taglib directive

```
<%@ directive attribute="value" %>
```

JSP Directives

1. JSP page directive

- The page directive defines attributes that apply to an entire JSP page.
- Syntax of JSP page directive:

```
<%@ page attribute="value" %>
```

Attributes of JSP page directive

- import, contentType, extends, info, buffer, language, isELIgnored, isThreadSafe, autoFlush, session, pageEncoding, errorPage, isErrorPage

JSP Directives

1)import

- Used to import class, interface or all the members of a package.
- Example of import attribute

```
<html>
<body>

<%@ page import="java.util.Date" %>
Today is: <%= new Date() %>

</body>
</html>
```

JSP Directives

2. JSP include directive

- Used to include the contents of any resource it may be jsp file, html file or text file.
- **Advantage:** Code Reusability

<%@ include file="resourceName" %>

```
<html>
<body>

<%@ include file="header.html" %>

</body>
</html>
```

JSP Directives

3. JSP taglib directive

- When we want to use other tag library in our JSP page. Such as JSTL or custom library.

<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="E" %>

```
<html>
<body>

<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="E" %>

<E:set var = "name" value = "Hello EIOV"></E:set>

<E:out value = "${name}"></E:out>

</body>
</html>
```

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